

SmartNPS360 JavaScript Bridge

Reference

Web ↔ native communication for the Android and iOS mobile app

Host: <https://smartnps360.com/>

Bridge handlers documented: 18

App version: 1.0.0+41

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Overview

The SmartNPS360 mobile app loads the web app inside an in-app WebView. JavaScript on the page can call native features (GPS, files, push, auth, clock-in) through a bridge. The native app can also push status updates and geolocation fixes back into the page.

All bridge calls use `window.flutter_inappwebview.callHandler(handlerName, payload)` and return a Promise. Injected scripts also expose higher-level helpers on `window.SmartNPS360` and `window.SmartNPSNativeAuth`.

NATIVE APP DETECTION

- User-Agent contains `SmartNPS360App`
- `window.SmartNPS360.isNativeApp()` returns `true` in the mobile shell

TRUSTED ORIGINS

Most handlers only accept calls when the current page host is `smartnps360.com`, `www.smartnps360.com`, or a subdomain of `smartnps360.com`. Untrusted callers receive:

```
{
  "ok": false,
  "error": { "code": "untrusted_origin", "message": "Untrusted origin" }
}
```

COMMON RESPONSE ENVELOPE

```
{ "ok": true, /* payload fields */ }
{ "ok": false, "error": { "code": "string", "message": "string" } }
```

INJECTED SCRIPTS (AT PAGE LOAD)

Script	Purpose
SmartNPS360 bridge	Auth helpers, push toggles, clock-in gate, theme, legacy location bridge
Geolocation override	Replaces <code>navigator.geolocation</code> with native GPS
iOS popover fix	Layout fix for notification UI on iOS (internal)

For web developers: Use `bindClockInGate` or `ensureCanClockIn()` before clock-in. Do not submit attendance directly on button click without the native background-location gate. See the clock-in integration example in `web_integration/clock_in_page.example.js`.

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1. Auth & Session

Web → Native

authEvent

WEB ENTRY POINTS

- SmartNPSNativeAuth.login(user, session)
- SmartNPSNativeAuth.setSession(session)
- SmartNPSNativeAuth.logout()

REQUEST

Field	Type	Required	Description
action or type	string	Yes	login, signup, register, session, or logout
user or profile	object	For login/signup	Officer profile object
session, auth, or tokens	object	For login/session	Token bundle (access token, refresh token, expiry fields)

EXAMPLE — LOGIN

```
{
  "action": "login",
  "user": { "id": 42, "name": "Officer Smith" },
  "session": { "accessToken": "eyJ...", "refreshToken": "..." }
}
```

RESPONSES

Action	Success	Failure
login / signup	{ "ok": true, "action": "login" }	invalid_args if user missing
session	{ "ok": true, "action": "session" }	invalid_args if session missing

Action	Success	Failure
logout	{ "ok": true, "action": "logout" }	logout_blocked_on_duty if officer is still on duty

NATIVE SIDE EFFECTS

On login/session: stores tokens, marks officer logged in, starts duty heartbeat, syncs push token. On logout: clears session, stops tracking, removes push token (blocked while on duty).

Web → Native loginWithSanctum

DESCRIPTION

Native Sanctum login. Also auto-triggered when a login form with employee_no and password fields is submitted.

REQUEST

```
{ "username": "EMP001", "password": "secret" }
```

username is sent to the API as employee_no.

RESPONSE

```
{ "ok": true, "hasToken": true }
{ "ok": false, "error": { "code": "request_failed", "message": "Sanctum login failed" } }
```

2. Location & Clock-in

Web → Native `getCurrentLocation`

WEB ENTRY POINTS

- `Direct callHandler('getCurrentLocation', payload)`
- `navigator.geolocation.getCurrentPosition` (injected override)
- `SmartNPSBridge.postMessage / SmartNPSAndroid.postMessage` with `request_current_location`

REQUEST OPTIONS

Field	Type	Default	Description
<code>timeout_ms</code>	number	12000	Clamped 5000–45000
<code>required_accuracy_meters</code>	number	50	Clamped 5–500
<code>for_clock_in</code> or <code>purpose</code>	bool / string	false	Enables clock-in disclosure and mock-location checks
<code>allow_foreground_only</code>	boolean	false	Skips background-location gate

SUCCESS RESPONSE

```
{
  "ok": true,
  "location": {
    "latitude": 24.8607,
    "longitude": 67.0011,
    "accuracy": 12.5,
    "altitude": 45.0,
    "speed": 0.0,
    "heading": 0.0,
    "timestampMs": 1752494400000,
    "timestamp": "2026-07-14T12:00:00.000Z",
    "isMocked": false,
    "isSimulatedBySoftware": false,
    "acceptedFromLiveStream": true,
    "isFreshLiveLocation": true,
    "isCachedLocation": false,
  }
}
```

```
"ageMsWhenAccepted": 0,  
"maxAllowedAccuracyMeters": 50,  
"timeoutMs": 12000  
}  
}
```

COMMON ERROR CODES

untrusted_origin, clock_in_gate_in_progress, location_services_disabled, background_location_required, disclosure_required, permission_denied, permission_denied_forever, mock_location, fresh_location_unavailable, get_location_failed

Web ↔ Native startLocationWatch

DESCRIPTION

Starts a native GPS stream. Native pushes updates via `__smartnps_native_geo_emit`.

REQUEST

```
{  
  "watchId": 900001,  
  "options": {  
    "required_accuracy_meters": 50,  
    "timeout_ms": 12000,  
    "interval_ms": 1000,  
    "for_clock_in": true,  
    "allow_foreground_only": false  
  }  
}
```

RESPONSE

```
{ "ok": true, "watchId": 900001, "intervalMs": 1000 }
```

Web → Native clearLocationWatch

REQUEST

```
{ "watchId": 900001 }
```


RESPONSE

```
{ "ok": true, "watchId": 900001 }
```

[Web → Native](#) prepareClockIn

WEB ENTRY POINT

SmartNPS360.ensureCanClockIn()

DESCRIPTION

Runs the store-safe clock-in gate: disclosure, permissions, background location readiness.

RESPONSE — READY

```
{ "ok": true, "canClockIn": true, "prepareInFlight": false }
```

RESPONSE — BLOCKED

```
{
  "ok": true,
  "canClockIn": false,
  "prepareInFlight": false,
  "reason": "background_location_required",
  "title": "Background location required for shift attendance",
  "message": "Enable background location (Always / Allow all the time) ..."
}
```

[Web → Native](#) getBackgroundLocationStatus

WEB ENTRY POINT

SmartNPS360.getBackgroundLocationStatus()

RESPONSE

```
{
  "ok": true,
  "backgroundReady": false,
  "phase": "foregroundOnly",
  "deniedReason": "background_location_required",
  "disclosureAccepted": true,
}
```

```
"serviceEnabled": true,  
"canClockIn": false,  
"prepareInFlight": false,  
"title": "Background location required...",  
"message": "Enable Always...",  
"platform": "android"  
}
```

phase: none | foregroundOnly | backgroundReady

3. Push Notifications

Web → Native `getPushNotificationToken`

RESPONSE

```
{ "ok": true, "token": "fcm_token_string_or_null" }
```

Web → Native `getPushNotificationStatus`

WEB ENTRY POINTS

`SmartNPS360.getPushNotificationStatus(), syncPushNotifications()`

RESPONSE

```
{
  "ok": true,
  "enabled": true,
  "permissionGranted": true,
  "hasToken": true
}
```

Web → Native `setPushNotificationsEnabled`

WEB ENTRY POINT

`SmartNPS360.setPushNotificationsEnabled(enabled)`

REQUEST

`true / false` or `{ "enabled": true }`

RESPONSE

Same shape as `getPushNotificationStatus`. May include `"unchanged": true`.

4. Files & Sharing

Web → Native pickFile

RESPONSE — SELECTED

```
{
  "ok": true,
  "canceled": false,
  "file": { "path": "/path/to/file.pdf", "name": "file.pdf", "size": 12345, "extension":
"pdf" }
}
```

RESPONSE — CANCELED

```
{ "ok": true, "canceled": true }
```

Web → Native pickImage

RESPONSE

```
{ "ok": true, "canceled": false, "image": { "path": "...", "name": "..." } }
```

Web → Native takePhoto

RESPONSE

```
{ "ok": true, "canceled": false, "photo": { "path": "...", "name": "..." } }
```

Web → Native downloadFile

REQUEST

URL string or { "url": "https://...", "filename": "report.pdf" }

RESPONSE

```
{ "ok": true, "file": { "path": "/app/documents/downloads/report.pdf", "name": "report.pdf" } }
```

Web → Native shareContent

REQUEST

Text string or { "text": "...", "subject": "..." }

RESPONSE

```
{ "ok": true }
```

5. Device & Navigation

Web → Native `getDeviceInfo`

RESPONSE (ANDROID EXAMPLE)

```
{
  "ok": true,
  "platform": "android",
  "device": { "brand": "Google", "manufacturer": "Google", "model": "Pixel 8", "sdkInt":
34 }
}
```

Web → Native `openExternalUrl`

REQUEST

URL string or { "url": "https://..." }

RESPONSE

```
{ "ok": true, "launched": true }
```

NAVIGATION (NATIVE, NOT A HANDLER)

Behavior	Description
Internal URLs	Loaded inside the WebView
External http(s) links	Opened in the system browser
Logout routes (/officer/logout, */logout)	Trigger native logout if officer is not on duty
Login routes	Pause native session and stop duty heartbeat
Push notification tap	Native app loads deep-link URL directly

6. Theme

Web → Native

themeChanged

WEB ENTRY POINTS

- `SmartNPS360.notifyTheme('dark')` or `notifyTheme('light')`
- Auto-detected via `DOM MutationObserver` on page theme classes

REQUEST

Boolean or { "isDark": true, "theme": "dark", "mode": "dark" }

RESPONSE

```
{ "ok": true, "isDark": true }
```

Updates native status bar and navigation bar colors.

7. Native → Web Events & Callbacks

CUSTOM DOM EVENTS

Event	Detail	When fired
smartnps360:background-location	getBackgroundLocationStatus result	Permission changes, page load
smartnps360:push-notifications	Push status object	Page load, app resume
smartnps360:push-token	{ "token": "fcm..." }	iOS after login / token sync

CALLBACK HOOKS ON WINDOW.SMARTNPS360

Callback	Argument
onBackgroundLocationStatusChanged(status)	Background location status object
onPushNotificationStatusChanged(status)	Push status object
onPushTokenReady(token)	FCM token string (iOS)

GEOLOCATION WATCH CALLBACKS

Function	Payload
window.__smartnps_native_geo_emit(watchId, position)	Standard Geolocation position object with native metadata
window.__smartnps_native_geo_error(watchId, err)	{ "code": 2, "message": "..." }

WATCH POSITION PAYLOAD

```
{
  "coords": {
    "latitude": 24.8607,
    "longitude": 67.0011,
    "accuracy": 12.5,
```



```
    "altitude": 45.0,  
    "altitudeAccuracy": null,  
    "heading": 0.0,  
    "speed": 0.0  
  },  
  "timestamp": 1752494400000,  
  "nativeSource": true,  
  "provider": "flutter_geolocator",  
  "is_mocked": false,  
  "is_simulated_by_software": false,  
  "accepted_from_live_stream": true,  
  "max_allowed_accuracy_meters": 50  
}
```

8. Web Helper API (window.SmartNPS360)

Injected at document start. All methods are no-ops or return rejected promises in a normal browser.

Method	Handler / behavior	Category
isNativeApp()	User-Agent check	Detection
notifyTheme(mode)	themeChanged	Theme
getBackgroundLocationStatus()	getBackgroundLocationStatus	Clock-in
ensureCanClockIn()	prepareClockIn	Clock-in
bindClockInGate(button, onAllowed, onBlocked)	Gate + status listener	Clock-in
getPushNotificationStatus()	getPushNotificationStatus	Push
setPushNotificationsEnabled(enabled)	setPushNotificationsEnabled	Push
syncPushNotifications()	Get status + broadcast	Push
broadcastPushNotificationStatus(status)	DOM event + UI sync	Push
bindPushNotificationToggle(input)	Auto-binds [data-smartnps-push-toggle]	Push
bindAlpinePushNotificationToggle(button)	Alpine.js toggle helper	Push

WINDOW.SMARTNPSNATIVEAUTH

Method	Sends to authEvent
login(user, session)	{ action: "login", user, session }
setSession(session)	{ action: "session", session }
logout()	{ action: "logout" }

9. Geolocation Override

Browser geolocation is disabled in WebView settings. The injected script replaces `navigator.geolocation` with native GPS.

Browser API	Native behavior
<code>getCurrentPosition(success, error, options)</code>	Runs clock-in gate (unless <code>allow_foreground_only</code>), then <code>getCurrentLocation</code>
<code>watchPosition(success, error, options)</code>	<code>startLocationWatch</code> + <code>__smartnps_native_geo_emit</code>
<code>clearWatch(id)</code>	<code>clearLocationWatch</code>

WEB OPTIONS → NATIVE OPTIONS

Web option	Native option
<code>enableHighAccuracy: false</code>	<code>required_accuracy_meters: 100</code>
<code>timeout</code>	<code>timeout_ms</code> (5000–45000)
<code>required_accuracy_meters</code>	Passed through (5–500)
<code>allow_foreground_only: true</code>	Skips clock-in gate
Clock-in gate unlocked	Sets <code>for_clock_in: true</code>

10. Legacy Location Bridge

For older web integrations that do not use the geolocation override.

Web → Native

SmartNPSBridge.postMessage / SmartNPSAndroid.postMessage

REQUEST

```
{ "action": "request_current_location" }
```

Native → Web

SmartNPSWeb.receiveLocation / receiveLocationError

SUCCESS CALLBACK PAYLOAD

```
{
  "latitude": 24.8607,
  "longitude": 67.0011,
  "accuracy": 12.5,
  "provider": "flutter_geolocator",
  "timestamp": 1752494400000,
  "timestamp_iso": "2026-07-14T12:00:00.000Z",
  "altitude": 45.0,
  "speed": 0.0,
  "heading": 0.0,
  "is_mocked": false
}
```

ERROR CALLBACK PAYLOAD

```
{ "code": "NATIVE_GPS_FAILED", "message": "..." }
```

Handler registration summary: pickFile, pickImage, takePhoto, getCurrentLocation, startLocationWatch, clearLocationWatch, getDeviceInfo, openExternalUrl, shareContent, downloadFile, getPushNotificationToken, getPushNotificationStatus, setPushNotificationsEnabled, getBackgroundLocationStatus, prepareClockIn, themeChanged, loginWithSanctum, authEvent.